

Macroeconomics I – Quiz (Fall, 2022)

Please read these rules and instructions before you start. If you violate following rules, **Don't Know** will not be accepted as an excuse.

- There are 2 parts in this quiz, **110** points in total, if you get higher than 100 then it would be count **100**.
 - You will get no point if you cheat.
 - Mathematical steps and explanations should clear state.
 - Quality of explanation determines the grade, so you should try your best to write down the economic intuition clearly.
 - Good luck!
-

Part I. Multiple Choice (*3 points for each; 30 points in total*)

1. All of the following are measures of GDP except the total:

A

- ☒ A. expenditures of all businesses in the economy
- ☒ B. income from all production in the economy
- ☒ C. expenditures on all final goods produced
- ☒ D. value of all final production

2. The market value of all final goods and services produced within an economy in a given period of time is called:

A

- A. gross domestic product.
- B. gross national product.
- C. the GDP deflator.
- D. general durable purchase.

3. If a \$1,000 increase in income leads to a \$750 increase in consumption expenditures, then the marginal propensity to consume is:

B

- A. 0.75 and the multiplier is $\frac{1}{3}$
- B. 0.75 and the multiplier is 4
- ☒ C. 0.25 and the multiplier is $\frac{1}{3}$
- ☒ D. 0.25 and the multiplier is 4

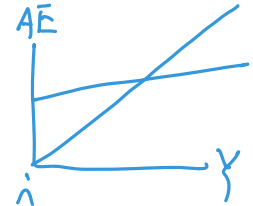
$$C = C_0 + C_1 Y_d$$
$$C = C_0 + C_1 (Y - T)$$
$$750 = 0.75 \times 1000$$
$$mpc = 0.75$$

4. Which of the following would tend to make the multiplier smaller?

B

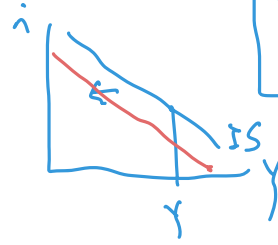
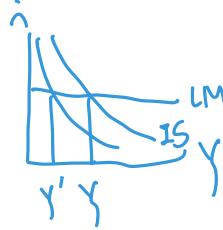
$$\frac{1}{1-C_1}$$

- ~~A.~~ an increase in the marginal propensity to consume.
- B. an increase in the marginal propensity to save.
- ~~C.~~ a reduction in taxes.
- ~~D.~~ a reduction in government spending.
- E. none of the above.



A 5. In the IS-LM model when taxation increase, in short-run equilibrium, in the usual case, the interest rate ____ and output ____

- A. falls; falls
- ~~B.~~ falls; rises
- C. rises; falls
- ~~D.~~ rises; rises



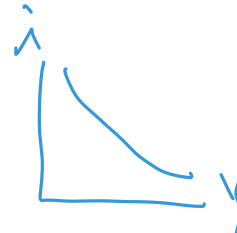
6. Which of the following generally occurs when a central bank pursues expansionary monetary policy?

- A. the central bank purchases bonds and the interest rate increases.
- B. the central bank purchases bonds and the interest rate decreases.
- C. the central bank sells bonds and the interest rate increases.
- D. the central bank sells bonds and the interest rate decreases.

C 7. We would expect which of the following to occur when the central bank pursues expansionary monetary policy?

- A. an increase in bond prices and an increase in the interest rate i
- B. a reduction in bond prices and an increase in i
- C. an increase in bond prices and a reduction in i
- D. a reduction in bond prices and a reduction in i
- E. none of these

$$\downarrow \hat{i} = \frac{100 - P_B \uparrow}{P_B}$$

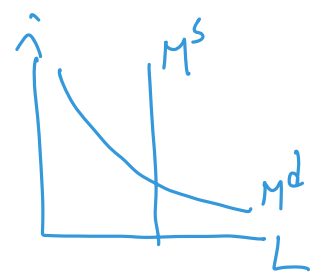
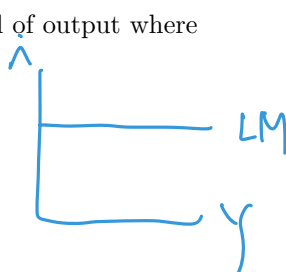


8. The IS curve represents

- A. the single level of output where the goods market is in equilibrium.
- B. the single level of output where financial markets are in equilibrium.
- C. the combinations of output and the interest rate where the money market is in equilibrium.
- D. the combinations of output and the interest rate where the goods market is in equilibrium.
- E. none of these

9. For each interest rate, the LM curve illustrates the level of output where

- A. the goods market is in equilibrium.
- B. inventory investment equals zero.



- C. money supply equals money demand.
- D. all of these
- E. none of these

10. An increase in government spending will likely have which of the following effects?

A

- A. a rightward shift in the IS curve
- B. a leftward shift in the IS curve
- ~~C. an upward shift in the LM curve~~
- ~~D. a downward shift in the LM curve~~

Part II. Problems and Short-essay Questions (70 points in total)

1. (12 points) During a given year, suppose the following activities occur in an economy.

- (a) An automobile manufacturing company pays its workers \$10 million to assemble 5,000 cars. The cars are then sold to an automobile store for \$12 million.
- (b) That year, the store pays \$1 million in wages to its salespeople, who sell the cars directly to consumers for \$15 million.

- i. Using the production-of-final-goods approach, what is GDP in this economy?
- ii. What is the value added at each stage of production? Using the value-added approach, what is GDP? $12 + (15 - 12) = 15$
- iii. What are the costs incurred in terms of wage payment and the profits earned? Using the income approach, what is GDP? $11 + 4 = 15$

2. (12 points) Suppose that an economy is characterized by the following behavioral equations (in billions of dollars)

$$C = 480 + 0.5Y_D$$

$$I = 110$$

$$T = 70$$

$$G = 250$$

Solve the following variables.

- (a) Equilibrium GDP (Y) $480 + 0.5(Y_D) + 110 + 250 - 70 = Y_D + 0.5Y_D$
- (b) Disposable income (Y_D)
- (c) Consumption spending (C)

3. (12 points) Use the economy described in 2.

- (a) Compute private saving, public saving, and investment spending.
- (b) Compute total demand.

private saving $S = Y - C - T = 1610 - 1250 - 70 = 290$

public saving $= T - G = 70 - 250 = -180$

$I = \text{private saving} + \text{public saving} = 290 - 180 = 110$

$$Y = C + I + G = 1610$$

(c) Assume that G is now equal to \$300 billion. Solve for equilibrium output, consumption, and disposable income. Why do you think the government will decide to expand fiscal spending?

4. (12 points) Suppose the money demand of an economy is given by

$$40000 \times 0.15$$

$$M^d = \$Y(0.25 - i)$$

$$M^d = M^s = 8000 = 40000(0.25 - i)$$

$$\hat{i} = 0.05 = 5\%$$

where $\$Y$ is \$40,000. Also, suppose that the supply of money is \$8,000.

(a) What is the equilibrium interest rate?

$$M^d = 6000$$

(b) Suppose the central bank increases the value of equilibrium interest rate, i , in part a to 10%, will there be excess money supply or money demand? What policy should the central bank consider to reach the new equilibrium interest rate?

contractionary OMO

5. (11 points) Use an IS-LM diagram to show the effects on output of a decrease in government spending.

Can you tell what happens to investment? why?

$$I = I(Y, \hat{i})$$

6. (21 points) Consider the following IS-LM model:

$$C = c_0 + c_1(Y - T)$$

$$I = b_0 + b_1Y - b_2i$$

$$Z = C + I + G$$

$$i = \bar{i}$$

(a) Solve for equilibrium output when the interest rate is $\bar{i}$. Assume $c_1 + b_1 < 1$.

(b) Solve for equilibrium level of investment, denoted I^* .

(c) Solve for equilibrium level of consumption, denoted C^* .